Active GLiNER: How to Mix Human and LLM Labels to Train a Small Extraction Model

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Abstract

Small extraction models such as GLiNER run on a CPU in milliseconds, but each new domain needs labelled training data. Human labels are slow and costly. A large language model (LLM) can label the data once instead, and the small model learns from its labels; on NER our student then runs 165 times faster than its teacher at almost the same F1. But LLM labels carry errors, so a practitioner must decide how to mix them with human labels: which examples the LLM should label, which LLM to use, how many labels to collect, and how the mix changes with the task. We answer these questions for one round of selection, with a GLiNER2.5 student, LLM teachers of two sizes and two model families (Gemma 4 E4B and 12B, GPT-6 Luna), and nine datasets covering NER, relations, multi-label classification, slot filling and two decision tasks. The main answer is that the mix depends on the task. Within two points of a student trained on human labels only, the LLM can label 50% or more of the inputs for decisions and 25% for NER, while for relations and slot filling even a quarter of LLM labels costs more: the LLM is weak on relations and fills slots the corpus leaves empty. To beat the LLM itself, a quarter or fewer human labels among 1,000 ranked inputs is enough on 6 of 7 original datasets with either teacher family. Picking the examples the student is least sure about loses to random picking on NER and multi-label classification, and on relations LLM label errors remove most of its gain. A larger LLM of the same family helps the student more than a second family does. We release the code as a command-line tool, together with all teacher labels and run logs.

1 Introduction

Information extraction models that run on a CPU in milliseconds are attractive for production, but they need labelled data for every new domain. Human annotation of that data is slow and costly. Calling a large language model (LLM) on every input at run time avoids training but is slow, costs money per call, and sends the data to a GPU server or an outside API. A middle path is to let the LLM label the training data once and train a small student on those labels (Zhang et al., 2023; Kholodna et al., 2024); such a student can match or beat its teacher at a fraction of the cost (Vedula et al., 2024). In our NER runs, a student trained on 400 Gemma-labelled sentences reaches 75.28 F1 against Gemma's 76.36 and runs 165 times faster (Section 4.5). GLiNER (Zaratiana et al., 2023) and GLiNER2 (Zaratiana et al., 2025) made such students general: one encoder handles entities, relations, classification and structured output from a schema. An LLM could also write the training texts themselves. The author's thesis tried this on MIT Movie: the larger the share of LLM-written texts with their labels, the lower the student's F1, and they could not replace human-labelled examples (Thomas, 2025). We therefore study LLM labels on real texts only.

If an LLM labels the training data, the next question is which data it should label. For human annotators the usual answer is active learning: let the current model pick the examples it is least sure about, because those should teach it the most. Yet active learning often fails to beat simply picking examples at random (Ghose and Nguyen, 2024; Yu and Fathima, 2026; Mittal et al., 2019), and LLM labels add a new risk. The examples a small model finds hardest may also be the ones an LLM gets wrong, so picking them could collect exactly the worst labels. Hakimi et al. (2026) even find that labelling the whole pool with an LLM beats active learning.

We ask the questions a practitioner faces, in the order they come up:

1. Which examples should the LLM label? (Section 4.1)

Question	Short answer
Why LLM labels?	One LLM pass labels the training data; the student then runs 165 times faster than the LLM at almost its F1 (§4.5).
Why not LLM-written texts?	In the thesis they could not replace human-labelled examples (Thomas, 2025).
Which examples?	Mostly not the ones the student is least sure about. Rank once with empty predictions last, then split the top at random (§4.1).
Which LLM?	A bigger one labels better on 5 of 6 datasets and trains a better student; another family adds 0.26 points to the student (§4.2).
How many labels?	About 1,000 ranked inputs. A quarter human beats the LLM on 6 of 7 datasets (§4.3).
What mix per task?	The LLM share that costs at most two points: decisions 50–100%, NER 25%, multi-label 0–25%, relations 0%, slots 0% (§4.4).

Table 1: Findings at a glance. Every number comes from the run logs.

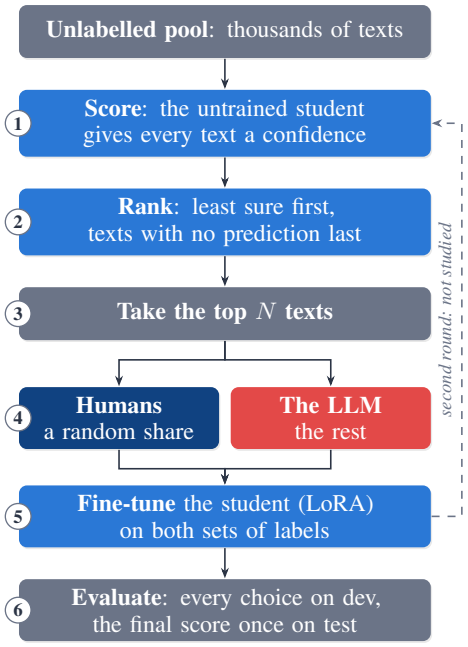


Figure 1: The one round we study. (1) The untrained student scores every text in the pool and (2) ranks them, least sure first. (3) The top N texts are labelled: (4) a random share by humans, the rest by the LLM. (5) The student is fine-tuned on both, and (6) evaluated. Section 4.1 compares step 2 with random picking, Section 4.2 changes the LLM, and Sections 4.3 and 4.4 vary N and the human share. A second round with the trained student (dashed) is not studied.

2. Which LLM: does a bigger one help, or one from another family? (Section 4.2)

3. How many labels, and how many of them from humans? (Section 4.3)

4. What mix of human and LLM labels does each task need? (Section 4.4)

5. Is it worth it against calling the LLM directly? (Section 4.5)

Table 1 gives the short answers, and Figure 1 shows the round we study, step by step. The main answer is that the mix depends on the task. Where the answer is one option from a short list, the LLM can label 50% or more of the inputs at little cost; for entity names, about a quarter; for relations and slot filling, every LLM label costs. A second answer holds across tasks: from 1,000 ranked inputs, a quarter or fewer human labels beats the LLM itself on 6 of 7 original datasets with either teacher.

We make five contributions.

1. A test of uncertainty-based selection against random selection, with human and with LLM labels, on four tasks and six datasets. We fixed the analysis before any run (preregistration), use five seeds on the main cells, and correct for running several tests at once.

2. A comparison of teachers: a larger and a smaller LLM from one family (Gemma 4 12B and E4B), and two families (Gemma and GPT-6 Luna), both as labellers and as sources of student training data (Section 4.2).

3. Mixing curves on every dataset with both teacher families: how many human labels it takes to beat the LLM, chosen on dev and confirmed on test (Section 4.3).

4. A cross-task guide on nine datasets and five task families, including two decision tasks: how many points each share of LLM labels costs, and which teacher errors explain it (Section 4.4).

5. An open command-line tool that runs the whole round (rank, select, label with any OpenAI-compatible endpoint, train, report), with every paper run reproducible through it. Its default human share follows the task type (Table 9).

This paper builds on the author's master thesis (Thomas, 2025), which studied the same round for NER on MIT Movie with GLiNER and Gemma 3.

Section 5 lists what the thesis did, what it found, and which of its protocol problems this study fixes.

2 Related work

Active learning with LLM annotators. LLMaAA (Zhang et al., 2023) uses an LLM as the annotator inside active learning for NER and relation extraction. Kholodna et al. (2024) report that GPT-4 labels in the loop are 42 times cheaper for low-resource NER. ALLabel (Chen et al., 2025) selects demonstrations for the LLM rather than training data for a student. Qi et al. (2026) combine several small LLMs and use their disagreement to flag errors, and HyPAC (Zeng et al., 2026) routes examples between fast and slow LLMs and humans with an error guarantee. Active knowledge distillation (Luccioli et al., 2025) sends uncertain examples to an LLM teacher for classification. We differ in the student (a schema-driven GLiNER2.5 model), the breadth of tasks, and the explicit test of whether teacher error cancels the gain from selection.

Doubts about active learning. Random selection is a strong baseline (Ghose and Nguyen, 2024; Mittal et al., 2019). For NER, Yu and Fathima (2026) find uncertainty and diversity rules unstable, and a scoping review finds that only 13 of 62 studies report run time (Kohl et al., 2024). Practitioners cite setup effort and unclear savings as blockers (Romberg et al., 2025). We therefore pre-register the contrast, report seeds and intervals, and release a tool.

LLM-to-small-model distillation. GLiNER-BioMed (Yazdani et al., 2025) and FiNERweb (Golde et al., 2025) train GLiNER-style students on LLM labels without selection. Vedula et al. (2024) show that a distilled BERT can beat its teacher. Merdjanovska et al. (2026) reweight noisy LLM NER labels by error type.

GLiNER and pre-annotation. The closest work to ours outside arXiv uses zero-shot GLiNER to pre-annotate hiking descriptions for human correction and retrains over three rounds (Tafer et al., 2026); it has no selection rule and no random comparison.

Uncertainty in extraction. Span probabilities can stay informative under domain shift (Zheng et al., 2026); calibration metrics for NER are discussed by Hashimoto et al. (2024). Classic active learning for NER uses sequence-level uncertainty such as MNLP (Shen et al., 2017).

3 Setup

Words we use. The *student* is the small model we train; the *teacher* is the LLM that labels data for it. The *pool* is the unlabelled training text the student picks from. *Dev* is a separate labelled set used for every choice we make, such as settings, early stopping and which cell to report; *test* is used once, for the final scores. *Ground truth* or *human labels* are the labels made by each dataset's annotators. *Zero-shot* means the student before any training. A *cell* is one setting: dataset, how the N training inputs are picked, and who labels them. A *seed* fixes the random choices of a run; we repeat each run with three or five seeds. We score with *micro-F1*, from 0 to 100: it counts both the correct labels the model misses and the wrong labels it adds. *LoRA* trains small add-on weights instead of the whole model.

3.1 Tasks and data

Table 2 lists the datasets. We chose sets that are small (at most about 15k training sentences), carefully annotated, and cover four output structures. After the main study we added three sets with open licences and human labels (lower block of Table 2): two decision tasks, where the answer is one option from a fixed list, and Universal NER on the English Web Treebank, an openly licensed counterpart to CleanCoNLL with person, organisation and location. They run the mixing curves of Section 4.4. BoolQ pairs a question with a Wikipedia passage and asks yes or no. CLINC150 asks for one of 150 intents or "out of scope". We also loaded SST-5 (five sentiment levels) but gave it no runs: its licence is not stated and its levels are crowd slider scores cut into bins.

For CrossRE we use an adapted protocol: entity mentions are given as input to every model, and the task is to predict all relation types for every ordered pair of mentions, including pairs with no relation. End-to-end triple extraction was too hard for both students and teachers to support a comparison (pilot micro-F1 below 0.08).

3.2 Student

The student is GLiNER2.5-multi (Fastino, 2026) (287M parameters, mDeBERTa-v3-base encoder), loaded from a pinned snapshot, trained with LoRA

Task	Dataset	Pool	Dev	Test	Labels
NER	CleanCoNLL (Rücker and Akbik, 2023; Tjong Kim Sang and De Meulder, 2003)	13,957	3,233	3,427	4 types
NER	BC5CDR (Li et al., 2016)	5,228	5,330	5,865	chemical, disease
NER	MIT Movie, corrected (Liu et al., 2013; Galileo, 2022)	8,797	978	2,443	12 types
Relations	CrossRE, 6 domains (Bassignana and Plank, 2022)	2,519	300	2,446	17 relation types
Multi-label	Hallmarks of Cancer (Baker et al., 2016)	12,119	1,798	3,547	10 hallmarks
Slots (JSON)	MASSIVE 1.1 en-US and fr-FR (FitzGerald et al., 2022)	11,514	2,033	2,974	55 slot types
NER	UNER EWT (Mayhew et al., 2024; Silveira et al., 2014)	12,543	2,001	2,077	3 types
Decision	BoolQ (Clark et al., 2019)	8,427	1,000	3,270	yes, no
Decision	CLINC150, plus (Larson et al., 2019)	15,250	3,100	5,500	150 intents, out of scope

Table 2: Datasets. MIT Movie has no official dev split: we hold out a seeded 10% of train. CrossRE pools train and dev and holds out 300 sentences as dev. BoolQ has no labelled test split: its validation split is our test set and a seeded 1,000 training questions are dev. Documents never cross splits in BC5CDR, CleanCoNLL, Hallmarks and UNER EWT. Sentences with no gold label stay in the pool. The lower block was added after the main study.

(Hu et al., 2021) through the `gliner2` trainer. Every run starts from a fresh copy of the base model. Hyperparameters come from one Optuna search per task on the dev set (20 trials, the first being the vendor-recommended recipe), frozen before any main run. Training stops after three dev checks without improvement, and the checkpoint with the best dev micro-F1 is kept. CrossRE is the exception: its dev score moves up and down between checks, so every CrossRE run trains all 1,000 steps and keeps its best dev checkpoint. For CrossRE the student scores each ordered mention pair with the checkpoint's pretrained relation scorer, applied to the given mention offsets, so that its zero-shot confidence is meaningful for selection. All runs use deterministic GPU settings in full 32-bit precision; even so, some runs differ slightly when repeated (Limitations), so we report the spread over seeds.

3.3 Teachers

The main teacher is Gemma 4 12B, a 4-bit version run on our own GPU with llama.cpp, at temperature 0 and with its output forced into a fixed JSON format. One prompt per task gives the label names and, where a dev check showed it helped, one short definition per label from the annotation guidelines. The same prompt is used for every teacher. A second teacher from another model family, OpenAI GPT-6 Luna, is called through the API at temperature 0 with reasoning turned off and the same prompt and JSON schema. It labels every pool, dev and test split, and trains the students of the main-table cells (min and random at the main budget) and of the ranked mixing curves, with 3 seeds. Teacher answers are validated: labels must be allowed and spans must occur exactly in the text, starting and ending on a word boundary of the student's tokeniser. A mention that occurs only inside a longer word (for example the rating "r" inside "rated") is dropped and counted as a teacher error. For a decision the teacher must return exactly one allowed option; any other answer trains nothing and counts as a miss, so a teacher's decision micro-F1 can differ slightly from its accuracy, while the student, which always answers, scores micro-F1 equal to accuracy. Teacher labels are cached, so every arm and seed sees the same labels for the same sentence.

3.4 Selection

From a frozen pool, a selector picks N sentences in one round. **Random** samples uniformly. **Min** scores every sentence with the zero-shot student and picks the lowest sentence confidence: the minimum span confidence for NER (no prediction counts as 0), the minimum distance of any label probability to the threshold for classification and relations, and for slots the minimum value confidence or, if nothing is predicted, one minus the highest below-threshold candidate. **Diversity** (NER only) clusters student sentence embeddings with k-means and takes the sentence nearest each centroid. Pool scores and embeddings are computed once and shared, so paired arms select identical sentences. Because "no prediction counts as 0" makes min prefer sentences where the student predicts nothing, a pre-registered sensitivity block on CleanCoNLL ranks such sentences last instead.

3.5 Protocol and statistics

Budgets are $N \in \{100, 400\}$ sentences (CrossRE $\{100, 200\}$). Primary cells (min and random, ground truth and Gemma, at the larger budget) have five seeds, all other cells three. Each seed sets pool tie-breaks, data order and LoRA initialisation. We report strict micro-F1 on the official test set; the test set is used once, for final

scores only. Two references bound every cell: a lower bound from four zero-shot GLiNER2 models (small, base and multi from GLiNER2.5, and GLiNER2 large), each with its threshold chosen on dev, and an upper bound from training the student on the whole pool, once with ground truth and once with Gemma labels.

The pre-registered primary contrast, per task, is the interaction

$$\Delta = (\text{min} - \text{random})_{\text{Gemma}} - (\text{min} - \text{random})_{\text{ground truth}},$$

paired by seed and, for NER, pooled with equal weight over the three datasets. In words: does picking the least-sure examples gain less over random picking when the LLM labels them than when humans do? A negative Δ means LLM errors erase part of the gain. The four interactions form one family of tests and the four practical contrasts $(\text{min} - \text{random})_{\text{Gemma}}$ a second; within each family we correct the p-values for running four tests at once (Holm). We report the seed spread and paired bootstrap intervals over test sentences (Dror et al., 2018).

4 Results

We answer the five questions of the introduction in order; each subsection opens with its short answer. All numbers come from 2,319 finished runs through `active-gliner analyse`. Intervals are 95% paired bootstrap intervals over test sentences; they do not include seed variation, so we give the seed standard deviation (SD) of each contrast as well.

4.1 Which examples should the LLM label?

Short answer. Mostly not the ones the student is least sure about. Picking them loses to random picking on NER and multi-label classification; on relations it wins with human labels, but LLM label errors remove most of the gain. For NER the least-sure sentences are often ones with nothing to find, and the ones the LLM labels worst. Ranking once with empty predictions last, then splitting the top at random between humans and the LLM, is what the rest of the paper uses.

Picking the least-sure examples mostly does not help. Min selection loses to random selection for NER and multi-label classification with both label sources; it helps for relations with human labels, and there LLM errors remove most of the gain. Table 3 gives every cell and Figure 2 the pre-registered contrast. The interaction Δ is 3.32 F1 points for NER (95% interval 2.89 to 3.80, Holm-adjusted $p < 0.001$, seed SD 2.93), -3.23 for relations (-4.80 to -1.91, $p < 0.001$, SD 2.97), -1.02 for classification (-2.87 to 0.75, $p = 0.512$) and 0.64 for slots (-0.47 to 1.78, $p = 0.512$). The practical contrast, min minus random with Gemma labels, is -6.96 for NER (-7.34 to -6.63, $p < 0.001$), 1.37 for relations (0.61 to 2.24, $p < 0.001$), -3.37 for classification (-4.48 to -2.32, $p < 0.001$) and 0.36 for slots (-0.37 to 1.10, $p = 0.390$).

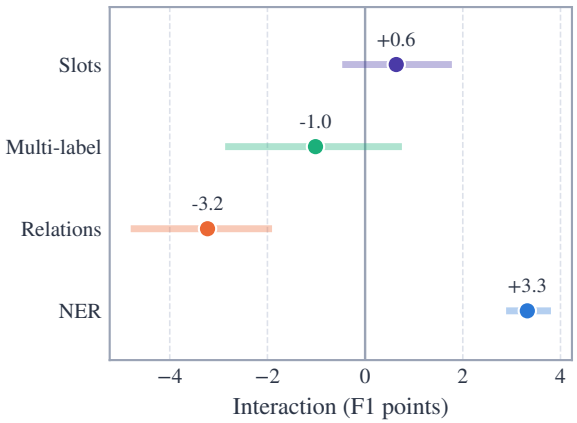


Figure 2: Primary interaction Δ per task, with 95% paired bootstrap intervals (bars; test uncertainty only). Positive: min loses less against random with Gemma labels than with ground truth.

The two contrasts must be read together. For NER, the interaction is positive because min loses to random under both label sources and loses less with Gemma labels; there is no gain for teacher error to erase. For relations, the pre-registered pattern appears: with ground truth, min beats random (37.80 against 33.20), and with Gemma labels most of that gain is gone (16.88 against 15.51). For classification, min loses under both sources. For slots, the selectors differ little, and the practical interval includes zero. The NER interaction varies across datasets: its seed SD is close to its mean, and on BC5CDR min loses about as much under both label sources, so the interaction there is near zero (Table 3).

On CleanCoNLL, the no-prediction sensitivity block separates two effects. With ground truth, min scores 70.67, min with empty sentences ranked last 84.46, and random 85.66. With Gemma labels the same three arms score 65.20, 75.07 and 72.24, and training on the whole

Dataset	Labels	Zero-shot	Random	Min	Min, empty last	Diversity	Whole pool
CleanCoNLL	Ground truth	55.7	85.7 ± 0.9 (5)	70.7 ± 4.1 (5)	84.5 ± 0.6 (3)	86.3 ± 0.9 (3)	89.6
	Gemma 4 12B		72.2 ± 2.0 (5)	65.2 ± 1.7 (5)	75.1 ± 0.3 (3)	70.6 ± 0.4 (3)	74.0
BC5CDR	Ground truth	64.7	82.5 ± 0.5 (5)	72.4 ± 2.2 (5)	83.9 ± 0.3 (3)	82.3 ± 0.5 (3)	87.4
	Gemma 4 12B		75.9 ± 0.7 (5)	65.4 ± 2.6 (5)	76.6 ± 1.8 (3)	75.7 ± 1.6 (3)	79.5
MIT Movie	Ground truth	55.5	84.2 ± 0.2 (5)	78.5 ± 0.7 (5)	84.8 ± 0.2 (3)	84.7 ± 0.7 (3)	88.4
	Gemma 4 12B		77.7 ± 1.7 (5)	74.4 ± 1.3 (5)	78.1 ± 0.5 (3)	78.0 ± 0.3 (3)	79.7
CrossRE	Ground truth	8.4	33.2 ± 2.2 (5)	37.8 ± 0.4 (5)	–	–	41.0
	Gemma 4 12B		15.5 ± 1.4 (5)	16.9 ± 0.4 (5)	–	–	12.5
Hallmarks	Ground truth	13.2	43.0 ± 3.3 (5)	40.6 ± 1.3 (5)	41.2 ± 1.8 (3)	–	42.3
	Gemma 4 12B		38.9 ± 1.5 (5)	35.6 ± 0.8 (5)	37.3 ± 0.8 (3)	–	40.5
MASSIVE en	Ground truth	31.6	70.0 ± 1.1 (5)	69.7 ± 0.7 (5)	70.7 ± 1.0 (3)	–	75.4
	Gemma 4 12B		48.5 ± 1.5 (5)	48.8 ± 1.3 (5)	49.9 ± 0.4 (3)	–	54.1
MASSIVE fr	Ground truth	26.2	62.0 ± 1.0 (3)	64.5 ± 0.9 (3)	64.0 ± 1.0 (3)	–	–
	Gemma 4 12B		46.7 ± 0.7 (3)	47.9 ± 0.4 (3)	47.2 ± 0.6 (3)	–	–

Table 3: Test micro-F1 (%) at the main budget ($N = 400$; CrossRE $N = 200$): mean ± seed SD (number of seeds). Zero-shot is the untrained student with a dev-chosen threshold. Whole pool is one seed. Generated by `active-gliner analyse`.

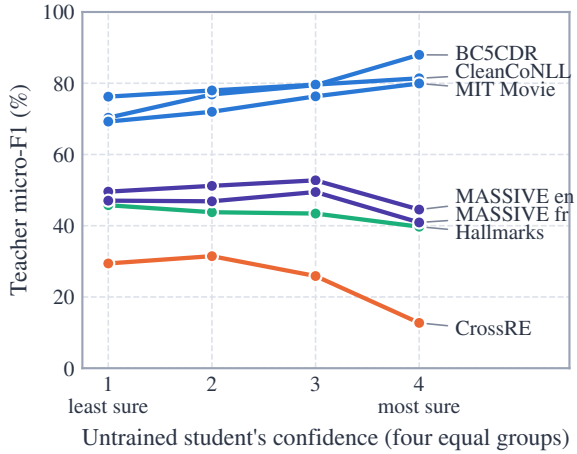


Figure 3: Gemma 4 12B micro-F1 on the pool, by the zero-shot student's sentence confidence (four equal-size groups). Colour: task type (blue NER, orange relations, green multi-label, violet slots).

Gemma-labelled pool scores 73.98. Most of the loss of min therefore comes from the rule that sentences with no prediction count as least confident: min then spends the budget on sentences where the student predicts nothing. With that rule reversed and Gemma labels, 400 selected sentences match or exceed the whole labelled pool on this dataset. This is one dataset and a sensitivity analysis, not a pre-registered contrast.

For NER, the LLM errs where the student is unsure. Gemma's F1 rises from the student's least sure quarter to its most sure quarter (Figure 3): 70.32 to 87.98 on BC5CDR, 69.24 to 79.94 on MIT Movie and 76.23 to 81.39 on CleanCoNLL. So uncertainty selection sends the teacher the sentences it labels worst. For the other tasks the pattern reverses at the top: the teacher is weakest where the student is most sure (CrossRE 29.41 against 12.72, Hallmarks 45.76 against 39.72, MASSIVE English 49.57 against 44.54). The bins alone do not predict the sign of the interaction: NER, where teacher errors follow student doubt, has a positive Δ, and relations, where they do not, has a negative one. The level of teacher error matters more: Gemma reaches only 22.62 F1 on CrossRE test, so any selected set carries many wrong labels.

The untrained student's confidence is meaningful. Ranking by the untrained student only makes sense if its confidence says something about its errors. It does, on every dataset (Figure 10, Table 10): the share of correct predictions rises with confidence. Among zero-shot predictions with confidence of 0.9 or more, 74.03% are correct on MIT Movie, 70.21% on BC5CDR and 64.69% on CleanCoNLL, against 20.95% in the lowest well-filled bin of CleanCoNLL (from 0.70). The level differs by task: 11.41% on CrossRE and 31.82% on Hallmarks, where the untrained student is confident and often wrong. Confidence orders sentences well enough to rank them, but it is not a calibrated probability before training.

4.2 Which LLM: a bigger one, or another family?

Short answer. A bigger one helps: Gemma 4 12B labels better than Gemma 4 E4B on 5 of 6 datasets and trains a better student. Another family helps less: GPT-6 Luna labels better than Gemma on 4 of seven datasets, but its students score only 0.26 points higher, because both families fail on many of the same sentences.

A bigger LLM labels better, and both sizes fail on the same examples. To test whether a bigger LLM helps within one family, we compare Gemma 4 12B with the smaller Gemma 4 E4B on frozen, stratified test samples (up to 1,000 sentences, whole documents where the corpus has them), weighting each sentence by its inclusion probability. The larger teacher is better on 5 of 6 English datasets: by 6.59 points on BC5CDR (4.15 to 9.10), 8.93 on CleanCoNLL (4.24 to 13.92), 12.85 on MIT Movie (10.45 to 15.26), 4.31 on Hallmarks (0.15 to 8.47) and 13.61 on MASSIVE (10.86 to 16.40). On CrossRE both fail alike (22.39 against 21.82; gap interval -1.85 to 2.97). The two teachers err on largely the same sentences: the overlap of their error sentences (shared errors divided by all sentences either gets wrong, the Jaccard overlap) is between 0.60 (MIT Movie) and 0.70 (CleanCoNLL), and 0.90 on CrossRE. The CleanCoNLL sample holds few documents, so its level estimate (65.16 for 12B) differs from the full test score (76.09); the paired gap uses the same sentences for both teachers and is not affected by this.

The bigger LLM trains a better student. Where we can test it, a better teacher gives a better student: with 400 random sentences, E4B labels train a student to 68.23 on CleanCoNLL and 35.82 on Hallmarks, against 72.24 and 38.95 with 12B labels (three seeds each).

Another family labels a little better, but the student gains little. GPT-6 Luna labels better than Gemma on 4 of the seven original test splits, by 2.61 F1 points on average (Table 4). It gains most on CrossRE (33.34 against 22.62) and loses on MIT Movie (68.78 against 73.95). The two families still fail on many of the same sentences: the Jaccard overlap of their error sentences is between 0.58 and 0.81, highest on CrossRE (0.81). This is a little below the overlap of the two Gemma models on the ladder samples, so many teacher errors come from the task and its annotation conventions, not from one model family. Better labels give only a slightly better student: with min selection, students trained on Luna labels score 0.26 points above students trained on Gemma labels, averaged over the seven datasets. The interaction of Section 4.1 is similar for both families: on the shared seeds, (min $-$ random) with teacher labels minus (min $-$ random) with ground truth averages 0.68 points for Luna and 0.88 for Gemma. These second-teacher numbers are descriptive and outside the Holm families.

4.3 How many labels, and how many from humans?

Short answer. About 1,000 ranked inputs, with a quarter or fewer from humans: that beats the LLM on 6 of 7 datasets with Gemma and on 6 with Luna. Beyond about 50 human labels, LLM labels on the rest of the pool stop helping.

This is the question the thesis asked on MIT Movie, now on every dataset. The student ranks the pool once, least confident first, with sentences where it predicts nothing ranked last: they stay in the pool, but Section 5 shows that ranking them first wastes the budget on them. It takes the top $N \in \{100, 400, 1{,}000, 2{,}500\}$ sentences ($N \leq 1{,}000$ on CrossRE, whose pool holds 2,519). A random share of 0, 25, 50, 75 or 100% of them gets human labels and the rest Gemma 4 12B labels. Three seeds per cell, the same N sentences for every share. Two references bound each panel of Figure 4: the student trained on the whole pool with Gemma labels (lower) and with human labels (upper), both with enough steps for two passes over the pool.

On the whole pool, the student lands near its teacher. Trained on the whole Gemma-labelled pool, the student lands near its teacher: 74.88 against 76.09 on CleanCoNLL and 78.73 against 79.04 on BC5CDR, and above it on MIT Movie (79.02 against 73.95) and MASSIVE French (50.58 against 45.56). Trained on the whole human-labelled pool, it reaches 89.98 on CleanCoNLL and 88.59 on MIT Movie. The upper reference is not a ceiling: on Hallmarks, 2,500 ranked human-labelled sentences (53.57) score above the whole pool (45.34), whose seeds spread widely.

A small human share beats the LLM. Table 5 gives, for each N, the smallest human share that beats Gemma. The share falls as N grows. On CleanCoNLL, 100 ranked sentences need half human labels (77.13 against 76.09), and 400 need a quarter (79.06). On MIT Movie and MASSIVE, Gemma labels alone are enough once the student sees enough ranked sentences (MIT Movie 76.42 with 100). Hallmarks is the exception: no share beats Gemma below 1,000 sentences. The last two columns choose on dev instead of reading the test grid: the cell with the fewest human labels whose

	Teacher F1		Error	Student, Min		Min − Random		
Dataset	Gemma	Luna	overlap	Gemma	Luna	Gemma	Luna	Human
CleanCoNLL	76.1	79.4	0.69	65.5	61.5	−5.6	−12.8	−16.0
BC5CDR	79.0	79.0	0.58	64.2	71.4	−11.5	−4.6	−10.2
MIT Movie	74.0	68.8	0.64	74.8	61.3	−2.9	−10.4	−6.1
CrossRE	22.6	33.3	0.81	16.8	24.6	0.7	4.7	4.2
Hallmarks	42.9	42.0	0.62	35.8	37.3	−3.2	−1.7	−2.4
MASSIVE en	48.7	55.9	0.65	48.5	54.5	−0.4	1.6	0.2
MASSIVE fr	45.6	48.7	0.65	47.9	44.8	1.1	0.1	2.5

Table 4: Two teacher families. Teacher F1 is test micro-F1 (%) of the labels on the full test split. Error overlap is the Jaccard overlap of the sentences each teacher gets wrong. Student columns are test micro-F1 at the main budget ($N = 400$; CrossRE $N = 200$), mean over the 3 seeds that all label sources share. Human is ground truth. Generated by `active-gliner analyse`.

		Whole pool		Human share to beat the LLM				Fewest (dev)	
Dataset	LLM	LLM	Human	100	400	1,000	2,500	Human+LLM	F1
CleanCoNLL	76.1	74.9	90.0	50%	25%	25%	25%	50+50	77.1
BC5CDR	79.0	78.7	87.4	100%	50%	25%	25%	250+750	79.9
MIT Movie	74.0	79.0	88.6	0%	0%	0%	0%	0+100	76.4
CrossRE	22.6	17.3	43.3	50%	25%	25%	–	75+25	30.4
Hallmarks	42.9	39.7	45.3	none	none	75%	50%	750+250	46.9
MASSIVE en	48.7	54.4	75.9	50%	0%	0%	0%	0+1000	51.7
MASSIVE fr	45.6	50.6	70.8	25%	0%	0%	0%	0+400	47.2

Table 5: Test micro-F1 (%). LLM: Gemma 4 12B on the test set. Whole pool: the student trained on every pool sentence with Gemma or human labels. Human share to beat the LLM: for N ranked sentences, the smallest human share whose mean test F1 is above the LLM (descriptive; "none": no share does, –: not run). Fewest (dev): the cell with the fewest human labels whose seed-mean dev F1 beats Gemma's dev F1 with a paired bootstrap lower bound above zero, as human + Gemma-labelled sentences, and its test F1.

dev F1 beats Gemma's, then one test comparison per dataset, Holm-corrected over the seven. On test the chosen cell beats Gemma on MIT Movie with no human labels (2.47 points, interval 1.31 to 3.70, $p < 0.001$) and on MASSIVE English with none (2.96, $p < 0.001$), on CrossRE with 75 human labels (7.75, $p < 0.001$), on BC5CDR with 250 (0.85, $p = 0.042$) and on Hallmarks with 750 (3.98, $p < 0.001$). Two dev choices are not confirmed: CleanCoNLL with 50 human labels (1.04, interval −0.33 to 2.50, $p = 0.164$) and MASSIVE French with none (1.67, $p = 0.052$). The dev rule stops at the first cell that clears the bar, which on these two is a cell just above the teacher; larger cells beat Gemma on test by more (Figure 4).

The result holds against a second LLM family. We repeat the ranked curves with GPT-6 Luna as the teacher (Table 6, Figure 7); the human-only cells are shared with the Gemma curves. The result holds. From 1,000 ranked sentences, a human share of a quarter or less beats Luna on 6 of 7 datasets, as it beats Gemma on 6. The dev-chosen cell with the fewest human labels beats Luna on test on 7 of 7 datasets after Holm correction, against 5 of 7 for Gemma. A stronger teacher needs more human labels. On MASSIVE English, where Luna labels better than Gemma (55.93 against 48.75), the fewest cell uses 50 human labels where Gemma needed 0. On CrossRE, where Luna gains most, the dev rule needs 1,000 human labels (9.35 points above Luna, $p < 0.001$). A weaker teacher is beaten without human labels: on MIT Movie, where Luna trails Gemma, 400 Luna-labelled ranked sentences beat Luna by 3.75 points ($p < 0.001$).

The ranking gains little over random selection. Against random selection of the same number of sentences, the ranking gains little on average and varies by dataset. With Gemma labels only, it helps most on CleanCoNLL (73.00 against 69.08 at 100 sentences); with human labels only, it helps on CrossRE (34.02 against 30.35) and hurts on Hallmarks (25.11 against 32.56).

Gemma labels add little to the same human labels. The mixed cells beat the LLM because Gemma labels are free, not because they add accuracy. With the same human-labelled sentences

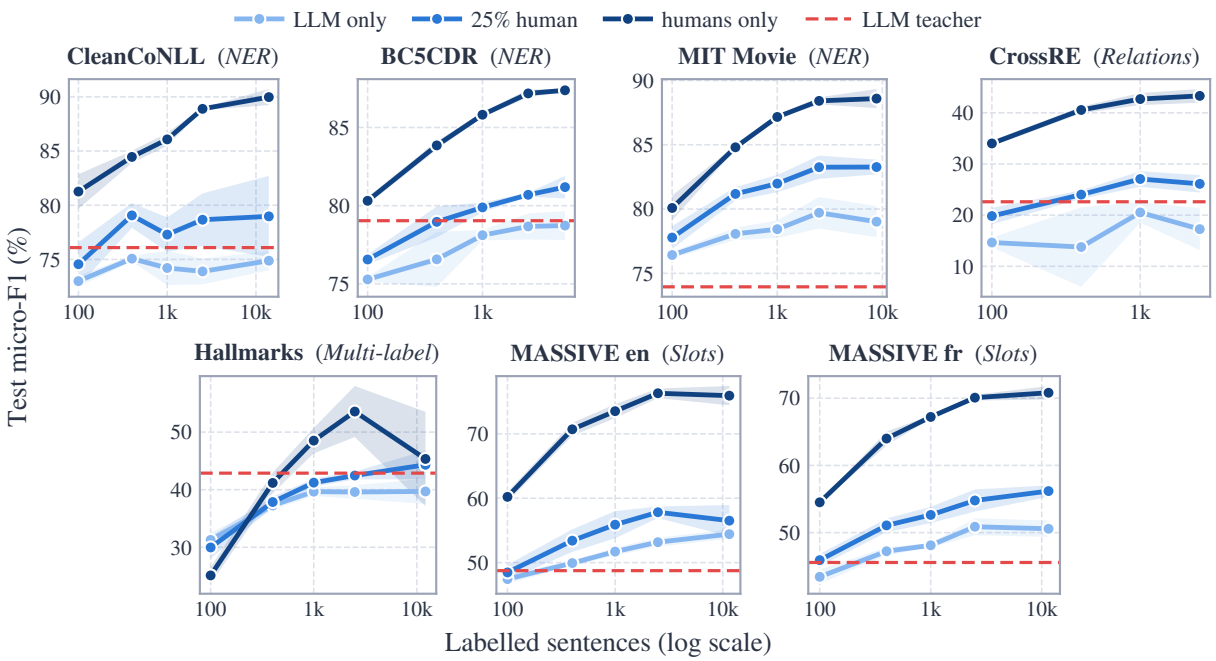


Figure 4: Test micro-F1 against the number of training sentences (log scale). Light blue: Gemma 4 12B labels only; blue: a quarter human labels and Gemma labels on the rest; dark blue: human labels only. The rightmost point of each line is the whole pool. Red dashed line: Gemma's own test F1. Shaded bands: SD over three seeds. The 50% and 75% shares lie between the blue lines and are left out for clarity; Table 5 uses all five shares.

		Whole pool		Human share to beat the LLM				Fewest (dev)	
Dataset	LLM	LLM	Human	100	400	1,000	2,500	Human+LLM	F1
CleanCoNLL	79.4	76.0	90.0	75%	25%	25%	25%	100+0	81.3
BC5CDR	79.0	79.9	87.4	75%	50%	25%	25%	200+200	79.9
MIT Movie	68.8	71.8	88.6	25%	0%	0%	0%	0+400	72.5
CrossRE	33.3	29.0	43.3	100%	50%	25%	–	1000+0	42.7
Hallmarks	42.0	42.9	45.3	none	none	50%	25%	750+250	45.0
MASSIVE en	55.9	53.6	75.9	50%	25%	25%	50%	50+50	57.9
MASSIVE fr	48.7	48.0	70.8	75%	25%	25%	25%	75+25	51.2

Table 6: The same as Table 5, with GPT-6 Luna as the teacher: LLM is Luna's test F1, the mixed cells have Luna labels for the non-human share, and Fewest (dev) beats Luna's dev F1. The human-only cells are shared with Table 5.

and nothing else, the student does as well or better: 100 random human-labelled CleanCoNLL sentences alone score 81.64. Adding Gemma labels for the whole rest of the pool to the same 100 human labels lowers this to 74.47 (Figure 6): thousands of teacher labels outvote a few hundred human ones. The pattern holds on all seven datasets: Gemma labels on the rest of the pool help only with very few human labels, 50 on most datasets and up to 200 on Hallmarks (with 50: 40.27 against 21.85), and lower the score beyond that. For a practitioner, the cheapest student that beats the LLM is a few hundred human labels on top-ranked sentences, with or without the LLM.

4.4 What mix does each task need?

Short answer. It depends on the task (Table 7). Within two points of all-human labels, the LLM can label 50–100% of the inputs for decisions, 25% for NER and 0–25% for multi-label classification. For relations and slot filling even a quarter costs more: the LLM is weak on relations (22.62 F1 for Gemma) and fills slots the corpus leaves empty.

The added sets run the ranked mixing curves of Section 4.3 with both teachers: $N \in \{100, 400, 1{,}000, 2{,}500\}$, human shares of 0 to 100%, random selection and whole-pool references, three seeds each (Figures 8 and 9). The decision tasks use the classification training set-

				Points lost when the LLM labels					
Task	Dataset	Teacher	T/S	25%	50%	75%	100%	Safe	Beat
NER	CleanCoNLL	Gemma	88%	1.5	5.9	8.8	11.9	25%	25%
NER	CleanCoNLL	Luna	92%	1.9	3.7	5.7	9.7	25%	25%
NER	BC5CDR	Gemma	92%	1.2	3.6	5.9	7.7	25%	25%
NER	BC5CDR	Luna	92%	1.5	3.9	6.1	8.3	25%	25%
NER	MIT Movie	Gemma	85%	0.5	2.2	5.2	8.7	25%	0%
NER	MIT Movie	Luna	79%	1.7	5.1	9.7	15.6	25%	0%
Relations	CrossRE	Gemma	53%	2.8	7.8	15.6	22.2	0%	25%
Relations	CrossRE	Luna	78%	2.2	4.6	8.7	13.2	0%	25%
Multi-label	Hallmarks	Gemma	88%	1.7	5.9	7.3	8.9	25%	75%
Multi-label	Hallmarks	Luna	87%	3.5	6.2	7.3	8.1	0%	50%
Slots	MASSIVE en	Gemma	66%	6.1	12.7	17.6	21.8	0%	0%
Slots	MASSIVE en	Luna	76%	5.8	11.0	15.2	19.8	0%	25%
Slots	MASSIVE fr	Gemma	68%	5.1	10.7	14.6	19.1	0%	0%
Slots	MASSIVE fr	Luna	72%	5.8	12.7	18.3	19.8	0%	25%
NER	UNER EWT	Gemma	100%	1.0	2.4	3.6	4.8	25%	none
NER	UNER EWT	Luna	97%	1.7	2.8	4.1	5.4	25%	75%
Decision	BoolQ	Gemma	135%	1.4	−0.9	0.9	−0.7	100%	none
Decision	BoolQ	Luna	116%	−0.7	0.6	2.3	2.3	50%	none
Decision	CLINC150	Gemma	110%	0.2	0.8	4.3	5.2	50%	none
Decision	CLINC150	Luna	116%	−1.0	−0.3	−0.1	0.5	100%	none

Table 7: How much LLM labelling each task tolerates, from 1,000 ranked inputs (test F1, seed means). T/S: the teacher's own test F1 divided by the F1 of the student trained on the same inputs with human labels; it compares a teacher with a student, not with human annotators. 25% to 100%: points lost against all human labels when the LLM labels that share of the inputs (negative: a gain). Safe: the largest LLM share, walking up from 25%, that loses at most two points. Beat: the smallest human share whose student beats the teacher (none: no share does). Descriptive; read from the test grid.

tings unchanged, without new tuning. BoolQ inputs (question plus passage) run up to 823 words, so its students update on 8 inputs in two micro-batches of 4. These results are descriptive: they were added after the pre-registered study, and their dev-chosen cells are tested separately from it.

The cost of LLM labels differs by task. Table 7 puts all nine datasets side by side, with MASSIVE in both languages. On decisions, half LLM labels cost at most 0.80 points, and on CLINC150 even all-Luna labels cost only 0.50. On the four NER sets, a quarter of LLM labels costs under two points with either teacher (CleanCoNLL 1.52, UNER EWT 1.04), and half costs more (CleanCoNLL 5.94). Relations and multi-label classification lose more per LLM label (CrossRE 7.76 at half with Gemma, Hallmarks 5.86). Slot extraction tolerates least: a quarter of Gemma labels costs 6.10 points in English and 5.07 in French.

Tolerating LLM labels is not beating the LLM. On decisions no human share beats the teacher, from any N we ran: at 1,000 ranked inputs a quarter or less of human labels beats it on 0 of 2 tasks, and no dev-chosen cell beats it on test. Gemma scores 86.25 on CLINC150 test and Luna 90.92, while the student trained on the whole human-labelled pool reaches 84.13. The gap is out-of-scope requests. On the 150 intents this student scores 91.29, above Gemma (87.04) and level with Luna (91.29), but it finds only 51.90% of out-of-scope requests, against 82.70% and 89.20%: they are 250 of 15,250 pool inputs and 1,000 of 5,500 test inputs. Larson et al. (2019) report the two scores apart; our single accuracy over 151 options mixes them. On BoolQ the student is the limit: with the whole human-labelled pool it scores 70.83, closer to always answering yes (62.17) than to Gemma (90.12). Reading a passage to answer a question asks more of a 287M encoder trained with LoRA than our extraction tasks do. On UNER EWT, Gemma labels as well as the all-human student (79.98 against 79.97); the dev-chosen cell that beats Luna has all 1,000 labels from humans (2.73 points on test, $p = 0.040$), although on the test grid three quarters already beat it (78.26 against 77.24). Where the teacher is far ahead of the student, LLM labels cost little because human labels add little.

The teacher's errors break corpus rules. Table 8 shows what goes wrong. The teacher prompts hold short label definitions, not the full annotation guidelines, so these errors are what an

Dataset	Teacher	Exact	Edges	Type	Both	Missed	Invented	Empty
CleanCoNLL	Gemma	76	3	16	2	4	2	5
BC5CDR	Gemma	80	9	0	1	10	12	7
MIT Movie	Gemma	69	7	3	3	18	5	18
UNER EWT	Gemma	85	7	2	1	5	19	7
MASSIVE en	Gemma	58	11	14	7	10	50	70
MASSIVE fr	Gemma	54	15	11	7	13	47	59
CleanCoNLL	Luna	79	2	15	1	3	2	6
BC5CDR	Luna	87	8	0	1	4	25	23
MIT Movie	Luna	69	22	2	3	4	4	64
UNER EWT	Luna	87	6	2	1	4	28	12
MASSIVE en	Luna	67	13	6	5	9	48	53
MASSIVE fr	Luna	58	20	5	5	12	47	50

Table 8: Teacher errors on test, in % of ground-truth spans: exact match, wrong edges (same type), wrong type (same edges), both wrong, missed. Invented: teacher spans that overlap no ground-truth span, in % of ground-truth spans. Empty: share of sentences without ground-truth spans to which the teacher adds one.

LLM does with a label name and one line of text; we read them descriptively. On MASSIVE English, Gemma matches only 58% of the ground-truth slots exactly, and it fills slots in 70% of the utterances that have none (Luna 53%), for example *alarm_type* on "tell me about my alarms". MASSIVE annotators marked only what an assistant needs to act on, with fixed rules for where a slot ends; Gemma marks much that looks like a slot. On Hallmarks the teacher gives a hallmark to 33% of the sentences the expert marked as none, at a label precision of 34%. On CleanCoNLL most errors are a wrong type (16% of mentions), and 79% of those are organisations labelled as locations: CoNLL marks a sports team named after its city as an organisation ("Karlsruhe 3 Freiburg 0"), and the teacher reads a place even though our organisation definition names teams. On UNER EWT, web text with the same three common types, Gemma matches 85% of the mentions exactly. These errors repeat across sentences, so a student trained on them can learn the teacher's habits and is scored on the corpus's. Across our tasks, the ones whose labels are one option from a list with plain names (decisions) or a few common entity types tolerate LLM labels best, and the ones whose labels also decide what counts, where it ends and which of many similar types applies (slots, multi-label topics, relations) tolerate them least. This is an association over nine datasets, not a tested cause.

4.5 Is it worth it?

Short answer. Yes, against calling the LLM on every input. On NER the trained student scores 75.28 F1 against its teacher's 76.36, runs 165 times faster, and cuts the estimated cost from 20.11 to 0.12 USD per million sentences.

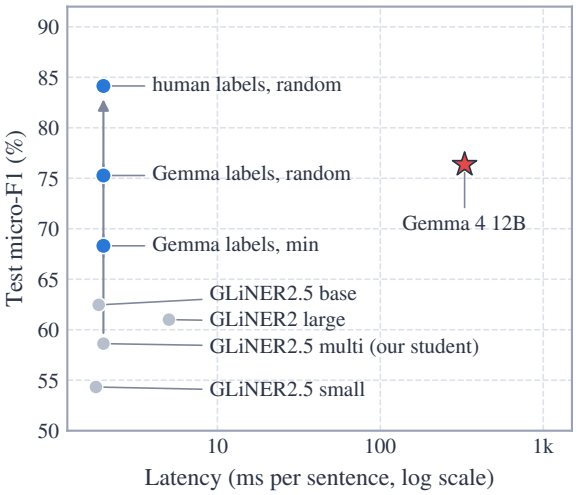


Figure 5: NER, pooled over the three datasets: test micro-F1 against latency on one RTX 3090. Grey: zero-shot GLiNER models; blue: our student fine-tuned on 400 sentences, by label source and selection rule; star: the Gemma 4 12B teacher. The arrow leads from the untrained student to its best fine-tuned version. Vertical bars: SD over seeds.

The student is much faster and cheaper than its teacher. Figure 5 places students and teacher on one chart for NER, pooled over the three datasets. Gemma 4 12B labels at 329.04 ms per sentence and reaches 76.36 F1. A student trained on 400 random Gemma-labelled sentences reaches 75.28 at 2.00 ms, 165 times faster; the estimated inference cost falls from 20.11 to 0.12 USD per million sentences. The same student trained on 400 ground-truth sentences reaches 84.14, and on 400 Gemma labels picked by min only 68.31; the untrained student scores 58.62.

The student can beat its teacher. Trained on the whole Gemma-labelled pool (one seed), it exceeds Gemma's own test F1 on MIT Movie (79.69 against 73.95), BC5CDR (79.54 against 79.04) and MASSIVE English (54.09 against 48.75); with only 400 random sentences it already does so on MIT Movie (77.68) and MASSIVE French (46.74 against 45.56). On CrossRE it stays far below the teacher (12.54 against 22.62), and the whole-pool CrossRE student trained on Gemma labels scores lower than the one trained on 200 random sentences.

5 Relation to the author's thesis

The thesis ran sixteen experiments on MIT Movie: zero-shot and fully fine-tuned GLiNER, an Optuna search and a LoRA layer study, confidence brackets and a threshold sweep, four selection scores over fourteen budgets, a hard-example comparison of GLiNER and Gemma 3, students on LLM versus ground-truth labels, a mixing grid of ground truth and LLM labels, four teachers, synthetic data, a prompt validator and a cost study. Its logs also show that Gemma 3's F1 drops from 69.4 on the full test set to 56.9 on the 100 training sentences where GLiNER was least confident; this study tests that observation on every dataset (Section 4.1).

The thesis protocol had problems that this study fixes: the MIT Movie dev file was a copy of the test file, so Optuna, early stopping and threshold choice used test data; the pool filter removed sentences with no ground-truth entity; the saved adapter was the last checkpoint rather than the best; the Optuna pruner never received intermediate scores; and no result was repeated with a second seed. We reran the thesis experiments on all seven dataset cells with the protocol above. One thesis conclusion does not hold: its three selection scores pick the same sentences, so their ranking was noise. Its central result, that a student trained on a mix of human and LLM labels beats the LLM with few human labels, holds on six of seven datasets (Section 4.3), with the caveat that the same human labels alone do about as well.

Thesis selectors measure noise. The thesis compared three sentence scores: mean confidence (avg), mean squared error from full confidence (MSE) and mean negative log-probability (MNLP). In all 63 cells (dataset, label source, budget and seed) the three scores select exactly the same sentences (63 of 63). Each score ranks a sentence with no prediction as least confident, and every pool holds more such sentences than the largest thesis budget: from 563 (6.40%) on MIT Movie to 10,545 (87.01%) on Hallmarks. The selected set is therefore a seeded random draw of empty sentences. For NER, min uses the same rule and selects the same set too (27 of 27 cells). Any score difference between avg, MSE and MNLP is training noise, which makes them a free repeatability test: with the same sentences and seeds, mean F1 differs by up to 4.02 points on Hallmarks, 1.59 on CrossRE and 1.46 on MASSIVE English, but only 0.00 on BC5CDR with ground truth. The thesis ranking of these scores cannot be read as a ranking of methods.

6 Conclusion: how to mix human and LLM labels

Which examples should the LLM label? In our one-round setting, mostly not the ones the student is least sure about. For NER those are the sentences the LLM labels worst (Section 4.1), and choosing them gains little over random selection once sentences with no prediction are ranked last: with Gemma labels on 400 sentences, CleanCoNLL reaches 75.07 against 72.24 for random and MIT Movie 78.07 against 77.68. The useful split runs the other way. The student ranks the pool once and takes its least confident sentences; a small random share of them gets human labels and the LLM labels the rest. We tested the obvious alternative only for NER: giving the hardest quarter of 400 sentences to humans, rather than a random quarter, scores 62.86 against 65.18 on CleanCoNLL, 59.49 against 64.87 on BC5CDR and 75.70 against 75.03 on MIT Movie (Gemma, empty predictions not ranked last, so the hardest quarter holds many sentences with no entity). From 1,000 ranked sentences, a quarter or less human labels beats the LLM on 6 of 7 datasets with Gemma and on 6 of 7 with GPT-6 Luna, and the cell with the fewest human labels, chosen on dev, holds on test on 5 of 7 datasets with Gemma and 7 of 7 with Luna. A stronger LLM raises the bar and needs more human labels. LLM labels matter most when human labels are almost absent: beyond about 50 human labels (200 on Hallmarks), LLM labels on the rest of the pool no longer help (Figure 6). The LLM is a free start, not a substitute for a few hundred human labels on the right

sentences.

How large a share the LLM can take depends on the task (Table 7). Where the answer is one option from a list with plain names, as in our decision tasks, LLM labels for half of the ranked inputs cost under a point, but the small student also stays below the LLM itself; there the choice is between the LLM and a cheaper, weaker student, and human labels change little. Where the labels are mentions of a few entity types, the LLM can take a quarter for under two points. Where the labels also encode what counts, where it ends and which of many similar types applies (multi-label topics, relations and above all slots), every LLM label costs, and many of the teacher's errors follow corpus rules that a short label definition does not carry (Table 8). A practitioner can look for this before training: score the LLM on a few hundred human-labelled dev inputs and count what it adds where the corpus marks nothing. With Gemma, the task with the most such additions, slot extraction, tolerates the fewest LLM labels; with Luna the signal is noisier (it adds entities to 64% of MIT Movie's few empty test sentences, yet MIT Movie tolerates LLM labels well). We offer this as a heuristic, not a tested rule.

A recipe. For a new extraction task with an LLM at hand:

1. Label a few hundred dev inputs by hand. Score the LLM on them and count how often it adds labels where the dev set has none.

2. Let the untrained student rank the unlabelled pool once, least sure first, with inputs where it predicts nothing ranked last.

3. Take the top 1,000 or so. Send a random share to human annotators, set by the task type (Table 9), and the rest to the LLM.

4. Train the student on both and compare it with the LLM on dev. If the LLM added many labels in step 1, raise the human share.

7 Limitations

Our results cover one round of selection, one student family and English plus French data. CrossRE uses given entity mentions, so its results do not cover end-to-end relation extraction. The hyphen-free tokenisation of the datasets hides a GLiNER2 word-splitting issue that matters for raw biomedical text. We cannot rule out that our models saw the evaluation data. No model owner states that our datasets were in training, and none states that they were removed. The GLiNER2 family reports zero-shot results on CrossNER, whose sentences and splits CrossRE reuses, so CrossRE test sentences were part of the student developers' entity evaluation; whether checkpoints were chosen on them is unknown. The student's training text includes news and PubMed, so the text (not the labels) of CleanCoNLL, BC5CDR and Hallmarks may overlap. CoNLL-2003 is a known contamination case for earlier LLMs, and public instruction collections contain CoNLL and BC5CDR tasks; our teachers do not document whether such data was used. MASSIVE postdates the student encoder's pre-training text. The full audit is in Appendix C. Training is not exactly repeatable on our GPU. NER runs on ground truth repeat almost to the digit (identical selections differ by at most 0.01 points), but relation, classification and slot runs, and NER runs on Gemma labels, can differ by up to 4.02 points on the same sentences and seed (Section 5). We therefore report the seed SD of every cell and of every contrast. Costs are estimates: GPU time at one rental price on one date, and ground-truth labels at a simulated crowd rate of 0.50 USD per sentence taken from a span-annotation study (Kasner et al., 2026). The planned API teachers (Qwen and gpt-oss on Cerebras) were not run; GPT-6 Luna replaces them as the second family, on the maintable cells and the ranked mixing curves with 3 seeds; the pre-registered Holm-tested interaction contrasts and the minimum-human-labels curve use Gemma alone. The smaller E4B teacher labelled only the ladder samples and two training

Task type	LLM share, ≤ 2 points	Human share (tool)
Decision	50–100%	50%
NER	25%	75%
Multi-label	0–25%	75%
Relations	0%	100%
Slots	0%	100%

Table 9: The mix per task type, from 1,000 ranked inputs (Table 7). LLM share: the range, over datasets and both teachers, of the largest LLM share that loses at most two points against all human labels. Human share: the tool's default, which keeps the loss within two points on every dataset of the type with Gemma labels. To beat the LLM only, a quarter human is enough on most datasets (Section 4.3).

cells.

The decision tasks and UNER EWT were added after the pre-registered study. They are descriptive, with their own Holm families, and the decision tasks use the classification training settings without new tuning, which may understate what the student can do on them. BoolQ students update on 8 inputs as two micro-batches of 4 to fit the GPU. This is not the same training as batches of 8: the length grouping of the sampler uses the physical batch, and batches of 8 drop the last partial batch of each epoch where micro-batches train it in a short extra update. Every BoolQ run uses this setup, so BoolQ arms compare with each other; 12 cells trained both ways differ by -0.50 points on average and by at most 1.83, within the repeat spread above. The T/S ratio of Table 7 divides by a student's score, so it reflects student capacity as well as teacher quality. SST-5 has no runs (Section 3), so no task with ordered levels is covered.

Three further points qualify single cells. The whole-pool Hallmarks run stopped early on a dev plateau, so it understates what the full pool can give; sentence-level Hallmarks is also much harder than the abstract-level task on which published scores are reported. Six CrossRE runs stopped with a non-finite gradient and one MASSIVE run made no optimiser step; all ran again with the same configuration and seed and finished normally. Runs killed by a GPU fault or out of memory were restarted in a new process with the same configuration and seed. The GLiNER2 trainer silently skips a batch that runs out of GPU memory; a check of epoch against step count in every training log found five runs with skipped batches. We made out of memory stop the run, and those five runs were repeated in full. Finally, fine-tuned students were not timed separately: they reuse the zero-shot timing of their base model, on the unmeasured assumption that a LoRA adapter adds little inference time.

Ethics and broader impact

All datasets are public research benchmarks. BC5CDR (annotations in the public domain; the PubMed abstracts may keep their publishers' copyright, so we release its labels as ids and offsets too), CrossRE and Hallmarks (GPL-3.0), MASSIVE (CC BY 4.0), UNER EWT (CC BY-SA 4.0), BoolQ (CC BY-SA 3.0) and CLINC150 (CC BY 3.0) carry open licences. Two do not: CleanCoNLL is built on Reuters RCV1 news text, which NIST distributes under a usage agreement, and its corrections ship without a licence file, and the corrected MIT Movie copy states no licence. We use both for research only and release their labels as sentence ids and character offsets, never the text. UNER EWT gives the NER results an openly licensed counterpart. No human annotators were involved. The GPU time of every run is in the released run logs.

Use of AI assistants

AI coding assistants wrote much of the code and the analysis scripts, reviewed each other's work, and drafted parts of the text. The author designed the study, directed every experiment, checked the results and every citation, and takes responsibility for the content.

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A Additional figures

Dataset	0.5	0.6	0.7	0.8	0.9
CleanCoNLL	–	–	21	27	65
BC5CDR	–	–	33	39	70
MIT Movie	–	–	32	41	74
CrossRE	6	7	10	10	11
Hallmarks	16	11	24	28	32
MASSIVE en	34	37	43	40	53
MASSIVE fr	35	29	34	41	47

Table 10: Zero-shot student: share of correct predictions (%) by confidence bin on the test set; – marks bins with fewer than 20 items.

B Training choices

Which weights to train. The default student trains LoRA adapters on the encoder and the task heads. With 400 random ground-truth sentences (three seeds), adapters on the encoder alone score within one point of the default on every dataset (MIT Movie 84.08 against 84.23, Hallmarks 43.67 against 42.98). Training the heads alone loses heavily everywhere (CleanCoNLL 65.14 against 85.66, MASSIVE English 50.07 against 69.98, BC5CDR 73.37 against 82.53) and collapses on Hallmarks (4.13). The encoder carries the adaptation. Full fine-tuning of all weights on the whole ground-truth pool (one seed, learning rate not tuned) scores below LoRA on the whole pool for BC5CDR (85.28 against 87.36), CleanCoNLL (88.79 against 89.63), MIT Movie (87.24 against 88.36) and MASSIVE English (71.88 against 75.36). It is higher only on Hallmarks (51.41 against 42.32), where the LoRA run stopped early. LoRA is enough for this student.

Thresholds and budgets. The thesis tuned the decision threshold on test data. On dev, the threshold matters for the zero-shot student and hardly at all after training: dev F1 moves by up to 11.92 points across thresholds 0.3 to 0.7 for the zero-shot student on Hallmarks (11.18 on MASSIVE English), against at most 1.99 for a trained student on any dataset.

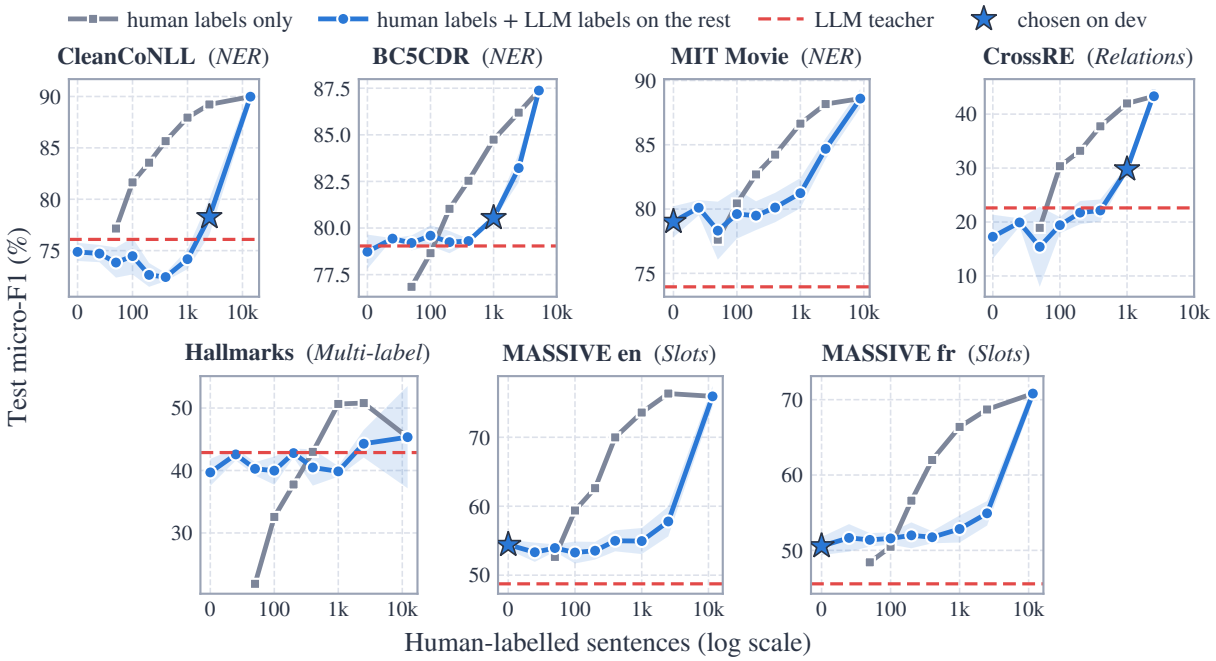


Figure 6: Same human labels, with and without Gemma labels on the rest of the pool. Blue: the whole pool, H sentences with human labels (a nested random set) and Gemma labels on all others; $H = 0$ is drawn at the left edge. Grey: the same number of random human-labelled sentences alone. Red dashed line: Gemma's test F1. Star: the smallest H chosen on dev (Section 4.3). Shaded bands: SD over three seeds.

C Data overlap audit

We checked papers, model cards and technical reports (September 2026) for each pair of evaluation dataset and model. We ran no probe, so nothing in this section is measured. Table 11 gives the result for the six datasets of the pre-registered study; the three sets added later (UNER EWT, BoolQ, CLINC150) are covered under Added sets below. No model owner states that any of our datasets was in training, and none states that it was removed.

Dataset	Student	Encoder	Teachers
CleanCoNLL	U	U	P, high
BC5CDR	U	P	P
MIT Movie	U	U	P, low
CrossRE	E	P	P, high
Hallmarks	U	P	P
MASSIVE	U	N	P

Table 11: Possible overlap of evaluation data with model training data. E: evaluated on by the model owner; P: a named path exists but no source confirms it; N: not possible by date; U: unknown. Student: GLiNER2.5-multi; encoder: mDeBERTa-v3 (CC-100); teachers: Gemma 4 E4B and 12B, GPT-6 Luna.

Student. The GLiNER2.5 release states that its training data was generated by an agent rather than assembled from existing datasets; it names no teacher, corpus or size and makes no decontamination statement (Fastino, 2026). It reports zero-shot results on five CrossNER domains. CrossRE annotates relations on the CrossNER sentences with the same splits (Bassignana and Plank, 2022), so the student's developers have scored entities on our CrossRE test sentences; whether they chose checkpoints on those scores is unknown. GLiNER2 trains on news, Wikipedia and PubMed text labelled by GPT-4o (Zaratiana et al., 2025), so the text, not the labels, of BC5CDR, Hallmarks and CrossRE may overlap with the zero-shot ladder models.

Encoder. mDeBERTa-v3 is pre-trained on CC-100, built from 2018 Common Crawl snapshots. MASSIVE (2022) and its source SLURP (2020) cannot be in it. Wikipedia and PubMed pages can be, so CrossRE, BC5CDR and Hallmarks text may be familiar to the encoder; a masked encoder sees text without our labels.

Teachers. Every dataset was public before the Gemma 4 data cutoff (January 2025). The Gemma 4 report states that benchmarks were decontaminated but does not list them. OpenAI does not publish the training data or a decontamination list for GPT-6 Luna, and all our test splits pre-

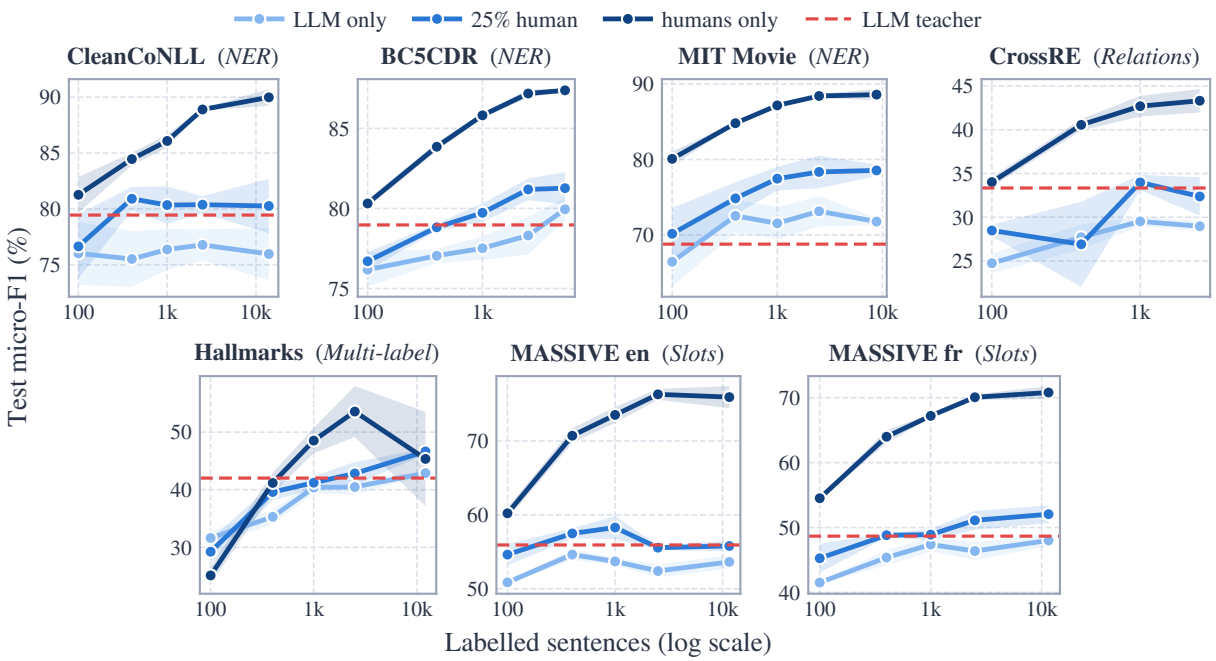


Figure 7: The mixing curves of Figure 4 with GPT-6 Luna as the teacher: the non-human share of each set has Luna labels. Red dashed line: Luna's own test F1. Shaded bands: SD over three seeds.

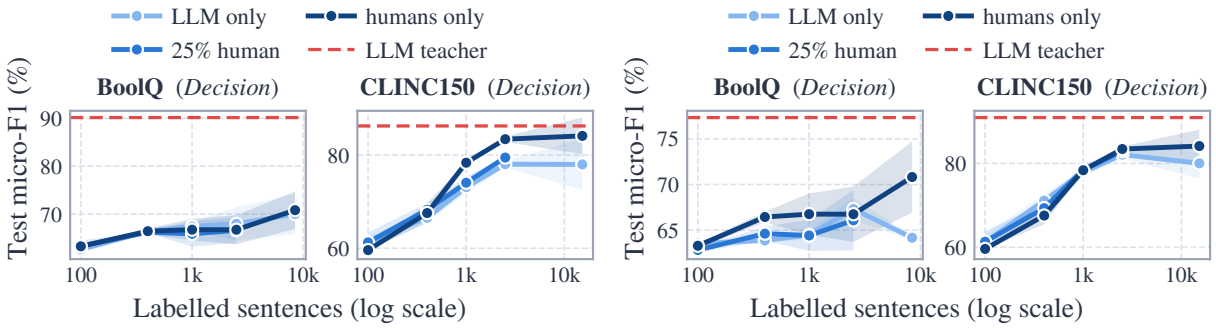


Figure 8: Mixing curves for the decision tasks (BoolQ, CLINC150), with Gemma (left) and Luna (right) labels for the non-human share. Red dashed line: the teacher's own test score. Shaded bands: SD over three seeds.

date it. CoNLL-2003 is a documented contamination case for ChatGPT, which reproduces its test sentences with labels. Public instruction collections contain CoNLL and BC5CDR tasks; no teacher owner says whether it used them. We therefore treat CleanCoNLL and the news domain of CrossRE, which is CoNLL-2003 text, as likely seen by the teachers.

Added sets. The GLiNER2.5 release lists CLINC-OOS among its zero-shot classification evaluations, so the student's developers have scored intents on our CLINC150 test inputs; whether they chose checkpoints on them is unknown. BoolQ is part of public instruction collections such as Flan (Longpre et al., 2023), so we treat it as likely seen by the teachers. UNER EWT is not among the GLiNER evaluations we found; its English Web Treebank text dates from 2012 and may be in pre-training text, without our labels.

Effect on our claims. Exposure raises absolute teacher and student scores. Our main claims compare selection arms, and teachers, on the same sentences. Pairing removes exposure that shifts every arm by the same amount, but exposure can still interact with label source or selection: a teacher that remembers test-like sentences labels them better, and that helps LLM-label arms more than ground-truth arms. It can also narrow the measured gap between teacher labels and ground truth. MASSIVE English and French are translations of the same utterances, so the two locales are not independent evidence.

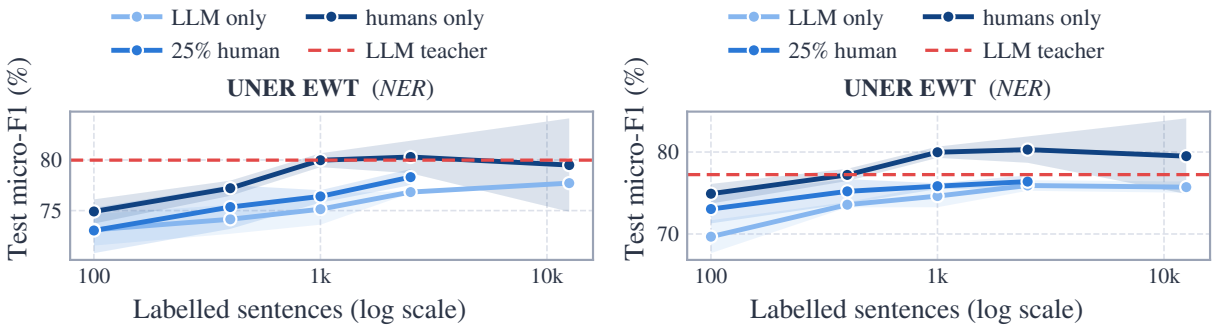


Figure 9: Mixing curves for UNER EWT with Gemma (left) and Luna (right) labels for the non-human share. Red dashed line: the teacher's own test F1. Shaded bands: SD over three seeds.

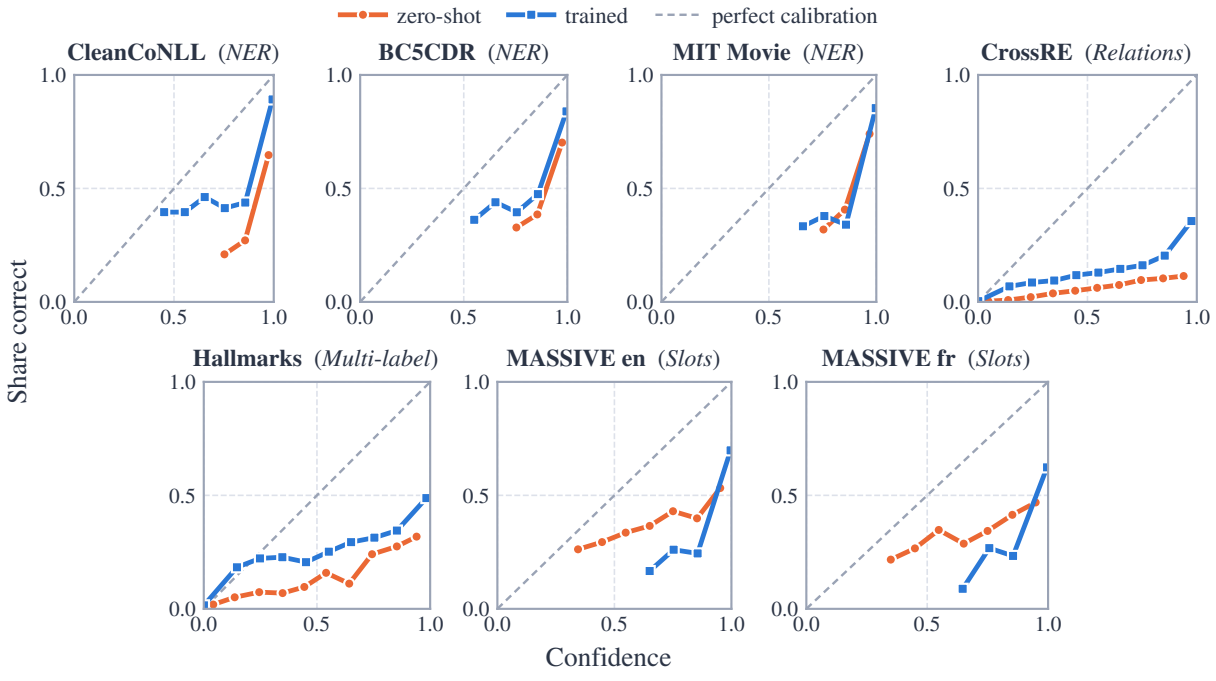


Figure 10: Share of correct predictions against confidence, test set, ten equal-width bins with at least 20 items; extraction counts emitted spans, classification and relations count every label decision. Orange: zero-shot student (the model that ranks the pool). Blue: student trained on 400 random human-labelled sentences (CrossRE 200), seeds pooled. Dashed: perfect calibration.